

Union Impact on Wage and Nonwage Outcomes

ECON 300 - Labour Economics

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- Discuss difficulties in measuring the impact of unions on members' wages and methods to overcome them.
- Describe mechanisms whereby unions may affect nonunion wages.
- Explain how unions affect wage inequality and the distribution of income.
- Evaluate how union impacts differ by skill, gender, sector, and product-market competition.
- Analyze effects on productivity, investment, profitability, and overall efficiency.

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 - Union increases wage in the organized sector, reduces employment there.
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Figure 15.1 – Two-Sector Model of General Equilibrium (*Placeholder*)

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Insert figure/diagram here later.

Additional Impacts and Threat Effects

- Nonunion firms may raise wages to compete for workers.
- **Threat effect:** nonunion employers raise wages to reduce risk of unionization.
- Threat effect stronger when:
 - Union wage is high
 - Nonunion sector easy to organize
 - Potential union aggressive
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- Typical union–nonunion differential:
 - U.S.: 10–20%
 - Canada: 10–25%
- **Longitudinal** (panel) estimates controlling for person fixed effects: $\sim 10\%$.

Empirical Evidence: Headline Findings

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Worked Example – Computing a Union Premium

- Nonunion pay structure (illustrative): $Y = 8000 + 2000S + 500X$.
- For a union worker with $S = 12$ and $X = 10$:

$$Y_U^* = 8000 + 2000(12) + 500(10) = \$37,000.$$

- Observed union earnings: $Y_U = \$44,000$.
- **Pure union premium:** $Y_U - Y_U^* = \$7,000$.

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Table: Worked Example Summary (Exact Values)

Quantity	Expression	Value
Expected nonunion earnings for union worker	$Y_U^* = 8000 + 2000(12) + 500(10)$	\$37,000
Observed union earnings	Y_U	\$44,000
Pure union earnings premium	$Y_U - Y_U^*$	\$7,000

- Use Blinder–Oaxaca to split average union–nonunion earnings difference into:
 - Part due to **endowments** (education, experience, occupation mix, etc.)
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Variation in the Union Wage Impact

- **By union density:** larger differentials where organization is widespread; often nonlinear.
- **By firm size:** positive relation with wages in both union and nonunion sectors; stronger for nonunion.
- **By industry and sector:** smaller in public sector; lower where foreign competition is stronger.
- **By skill and gender:** highest among low-skilled; union and nonunion wages similarly responsive to gender.
- **Over cycle/time:** union wages exhibit less cyclical variation.

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Figure 15.2 – Relationship between union and nonunion wages and personal/job characteristics.

Insert figure/diagram here later.

- Less wage dispersion among union workers in Canada.
- Larger impact on blue-collar wages.
- Net effect tends to **reduce** overall wage dispersion and inequality.

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Figure 15.3 – Wage dispersion comparison between union and nonunion sectors.

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- Wage and employment changes affect allocation of labour and capital.
- Potential **deadweight losses** if workers shift to lower-productivity nonunion jobs.
- Magnitude depends on:
 - Labour supply elasticity
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Figure 15.4 – Areas of deadweight loss under two-sector model with union wage.

Insert figure/diagram here later.

Fringe Benefits and Total Compensation

- Nonwage benefits are a higher share of compensation in union firms.
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- Unions can increase capital intensity and select more productive labour.
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- Identification requires rich controls; panel methods shrink differentials toward $\sim 10\%$.
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